

Gastrointestinal absorption of aluminium from single doses of aluminium containing antacids in man.



Ten subjects with normal renal function were given different single doses of aluminium containing antacids (1, 4, or 8 tablets). The antacid tablets (aluminium content 244 mg tablet⁻¹) were chewed and swallowed either with water, with orange juice, or with citric acid solution. There was a marked increase in serum concentration of aluminium when the antacids was ingested with citric acid (P less than 0.001) or with orange juice (P less than 0.05). When antacids were taken with water, a slight, but significant increase in serum aluminium concentration was seen with 4, but not with 1 or with 8 tablets. Following all doses of antacid, a significant increase in 24 h urinary excretion of aluminium was seen. The estimated absorption of aluminium was 8 and 50 times higher when antacids were taken with orange juice or with citric acid, respectively, than when taken with water. Thus, measurable quantities of aluminium are absorbed from single oral doses of antacids. The absorption is substantially enhanced by concomitant ingestion of citric acid.